

一、填空题(共 6 题, 每题 3 分, 共 18 分)

1. -27 ; 2. $\frac{A+2I}{4}$; 3. 24 ; 4. $\begin{pmatrix} 0 & 0 & -1 \\ 0 & 2 & 0 \\ 6 & 0 & 5 \end{pmatrix}$; 5. $\begin{pmatrix} 0 \\ -1 \end{pmatrix}$; 6. 4

二、选择题(共 6 题, 每题 3 分, 共 18 分)

D A C D A D

三、计算题(共 4 题, 共 28 分)

1.(6 分)

解: $|A| = \begin{vmatrix} 2 & 0 & 0 & \cdots & 0 & 1 \\ 0 & 2 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 2 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 2 & 0 \\ 1 & 0 & 0 & \cdots & 0 & 2 \end{vmatrix}$

$$= 2 \cdot (-1)^{1+1} \begin{vmatrix} 2 & 0 & \cdots & 0 & 0 \\ 0 & 2 & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 2 & 0 \\ 0 & 0 & \cdots & 0 & 2 \end{vmatrix} + 1 \cdot (-1)^{1+n} \begin{vmatrix} 0 & 2 & 0 & \cdots & 0 \\ 0 & 0 & 2 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 2 \\ 1 & 0 & 0 & \cdots & 0 \end{vmatrix}$$

$$= 2 \cdot 2^{n-1} + (-1)^{1+n} \cdot 1 \cdot (-1)^{n-1+1} \begin{vmatrix} 2 & 0 & \cdots & 0 \\ 0 & 2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 2 \end{vmatrix} = 2^n - 2^{n-2}$$

则 $\left| \frac{1}{2}A \right| = \left(\frac{1}{2} \right)^n |A| = \frac{1}{2^n} (2^n - 2^{n-2}) = \frac{3}{4}$.

2. (8 分)

解: 记矩阵 $A = (\alpha_1, \alpha_2, \alpha_3, \alpha_4, \alpha_5) = \begin{pmatrix} 1 & 1 & 2 & 2 & 1 \\ 0 & 1 & 1 & 5 & -1 \\ 1 & 0 & 1 & -3 & 2 \\ 1 & 1 & 0 & 4 & -1 \end{pmatrix}$

$$\xrightarrow{\text{初等行变换}} \begin{pmatrix} 1 & 0 & 0 & -2 & 1 \\ 0 & 1 & 0 & 6 & -2 \\ 0 & 0 & 1 & -1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} = (\xi_1, \xi_2, \xi_3, \xi_4, \xi_5) = U, \text{ 则}$$

1) 秩 $\{\alpha_1, \alpha_2, \alpha_3, \alpha_4, \alpha_5\} = 3$;

2) $\alpha_1, \alpha_2, \alpha_3$ 是向量组 $\alpha_1, \alpha_2, \alpha_3, \alpha_4, \alpha_5$ 的一个极大线性无关组;

3) $\alpha_4 = (-2)\alpha_1 + 6\alpha_2 + (-1)\alpha_3 = -2\alpha_1 + 6\alpha_2 - \alpha_3$,

$\alpha_5 = 1\alpha_1 + (-2)\alpha_2 + 1\alpha_3 = \alpha_1 - 2\alpha_2 + \alpha_3$.

3. (8 分)

解: $AB = A + 2B \Rightarrow (A - 2I)B = A$, 且 $A - 2I = \begin{pmatrix} 1 & 0 & 1 \\ 1 & -1 & 0 \\ 0 & 1 & 2 \end{pmatrix}$

$$(A - 2I, A) = \left(\begin{pmatrix} 1 & 0 & 1 & 3 & 0 & 1 \\ 1 & -1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 2 & 0 & 1 & 4 \end{pmatrix} \right) \xrightarrow{\text{初等行变换}} \left(\begin{pmatrix} 1 & 0 & 0 & 5 & -2 & -2 \\ 0 & 1 & 0 & 4 & -3 & -2 \\ 0 & 0 & 1 & -2 & 2 & 3 \end{pmatrix} \right) = (I, B)$$

$$\text{则 } B = \begin{pmatrix} 5 & -2 & -2 \\ 4 & -3 & -2 \\ -2 & 2 & 3 \end{pmatrix}.$$

4. (6 分)

$$\text{解: 矩阵 } A \text{ 的特征多项式 } |\lambda I - A| = \begin{vmatrix} \lambda - 4 & 3 & -5 \\ 0 & \lambda & 0 \\ 0 & -a & \lambda \end{vmatrix} = \lambda^2(\lambda - 4)$$

则 A 的特征值为 $\lambda_1 = \lambda_2 = 0, \lambda_3 = 4$;

A 可对角化, 则对 2 重特征值 $\lambda_1 = \lambda_2 = 0$, 齐次线性方程组 $(0I - A)x = 0$,

即 $Ax = 0$ 的基础解系包含的向量个数为 $2 = 3 - r(A) \Rightarrow r(A) = 1$, 从而 $a = 0$.

四、证明题(共 1 题, 8 分)

证: 由题意知齐次线性方程组 $Ax = 0$ 的基础解系含有 3 个线性无关的解向量.

显然, $\alpha_1 + \alpha_2, \alpha_2 + \alpha_3, \alpha_3 + \alpha_1$ 也是方程组 $Ax = 0$ 的解; 下证其线性无关.

不妨设 $k_1(\alpha_1 + \alpha_2) + k_2(\alpha_2 + \alpha_3) + k_3(\alpha_3 + \alpha_1) = 0$

整理得 $(k_1 + k_3)\alpha_1 + (k_1 + k_2)\alpha_2 + (k_2 + k_3)\alpha_3 = 0$

已知 $\alpha_1, \alpha_2, \alpha_3$ 是 $Ax = 0$ 的一个基础解系, 则 $\alpha_1, \alpha_2, \alpha_3$ 线性无关

$$\text{因此得 } \begin{cases} k_1 + k_3 = 0 \\ k_1 + k_2 = 0 \\ k_2 + k_3 = 0 \end{cases}, \text{ 有唯一解 } k_1 = k_2 = k_3 = 0$$

所以 $\alpha_1 + \alpha_2, \alpha_2 + \alpha_3, \alpha_3 + \alpha_1$ 线性无关, 也是方程组 $Ax = 0$ 的一个基础解系.

五、解方程组 (共 1 题, 14 分)

解: 增广矩阵

$$(A, b) = \left(\begin{pmatrix} -1 & -4 & 1 \\ 0 & a & -3 \\ 1 & 3 & a+1 \end{pmatrix} \middle| \begin{pmatrix} 1 \\ 3 \\ 0 \end{pmatrix} \right) \xrightarrow{\text{初等行变换}} \left(\begin{pmatrix} 1 & 4 & -1 \\ 0 & -1 & a+2 \\ 0 & 0 & (a+3)(a-1) \end{pmatrix} \middle| \begin{pmatrix} -1 \\ 1 \\ a+3 \end{pmatrix} \right)$$

①当 $a \neq -3$ 且 $a \neq 1$ 时, 方程组有唯一解;

②当 $a = 1$ 时, 方程组出现矛盾方程, 故方程组无解;

③当 $a = -3$ 时, 增广矩阵

$$(A, b) = \left(\begin{pmatrix} -1 & -4 & 1 \\ 0 & -3 & -3 \\ 1 & 3 & -2 \end{pmatrix} \middle| \begin{pmatrix} 1 \\ 3 \\ 0 \end{pmatrix} \right) \xrightarrow{\text{初等行变换}} \left(\begin{pmatrix} 1 & 0 & -5 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix} \middle| \begin{pmatrix} 3 \\ -1 \\ 0 \end{pmatrix} \right) = (U, d)$$

取 x_3 为自由未知量,

1) 令 $x_3 = 0$, 代入 $Ux = d$, 得原方程组的一个特解 $x_0 = (3, -1, 0)^T$;

2) 令 $x_3 = 1$, 代入 $Ux = 0$, 得 $Ax = 0$ 的一个基础解系 $\xi = (5, -1, 1)^T$;

则原方程组的通解为 $x = x_0 + k\xi = \begin{pmatrix} 3 \\ -1 \\ 0 \end{pmatrix} + k \begin{pmatrix} 5 \\ -1 \\ 1 \end{pmatrix}$, k 任意.

六、二次型 (共 1 题, 14 分)

$$\text{解: 二次型对应的矩阵 } A = \begin{pmatrix} 1 & -1 & 1 \\ -1 & a & -1 \\ 1 & -1 & 1 \end{pmatrix},$$

(1) 因二次型的秩为 1, 即 $r(A) = 1$, 则 $|A| = 0$, 得 $a = 1$;

$$(2) A = \begin{pmatrix} 1 & -1 & 1 \\ -1 & 1 & -1 \\ 1 & -1 & 1 \end{pmatrix}, \text{ 其特征多项式 } |\lambda I - A| = \begin{vmatrix} \lambda - 1 & 1 & -1 \\ 1 & \lambda - 1 & 1 \\ -1 & 1 & \lambda - 1 \end{vmatrix} = \lambda^2(\lambda - 3)$$

则 A 的特征值为 $\lambda_1 = \lambda_2 = 0, \lambda_3 = 3$.

①对于特征值 $\lambda_1 = \lambda_2 = 0$, 由 $(\lambda_1 I - A)x = 0$ 即 $Ax = 0$,

$$\text{即 } \begin{pmatrix} 1 & -1 & 1 \\ -1 & 1 & -1 \\ 1 & -1 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 0, \text{ 得基础解系 } \begin{cases} \xi_1 = (1, 1, 0)^T \\ \xi_2 = (-1, 0, 1)^T \end{cases}$$

1) 正交化: 取 $\beta_1 = \xi_1 = (1, 1, 0)^T$,

$$\text{令 } \beta_2 = \xi_2 - \frac{(\xi_2, \beta_1)}{(\beta_1, \beta_1)} \beta_1 = (-\frac{1}{2}, \frac{1}{2}, 1)^T,$$

2) 单位化: 令 $\eta_1 = \frac{1}{\|\beta_1\|} \beta_1 = (\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0)^T$;

$$\eta_2 = \frac{1}{\|\beta_2\|} \beta_2 = (\frac{-1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}})^T;$$

②对于特征值 $\lambda_3 = 3$, 由 $(\lambda_3 I - A)x = 0$

$$\text{即 } \begin{pmatrix} 2 & 1 & -1 \\ 1 & 2 & 1 \\ -1 & 1 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 0, \text{ 得基础解系为 } \xi_3 = (1, -1, 1)^T,$$

单位化得: $\eta_3 = \frac{1}{\|\xi_3\|} \xi_3 = (\frac{1}{\sqrt{3}}, \frac{-1}{\sqrt{3}}, \frac{1}{\sqrt{3}})^T$;

③ 记矩阵 $Q = (\eta_1, \eta_2, \eta_3) = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & \frac{-1}{\sqrt{3}} \\ 0 & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{3}} \end{pmatrix}$, 则 Q 为正交矩阵,

$$\text{且使得 } Q^T A Q = Q^{-1} A Q = \Lambda = \begin{pmatrix} 0 & & \\ & 0 & \\ & & 3 \end{pmatrix};$$

④令 $x = (x_1, x_2, x_3)^T$, $y = (y_1, y_2, y_3)^T$, 做正交变换 $x = Qy$,

原二次型就化为 标准形 $x^T A x = y^T (Q^T A Q) y = 3y_3^2$.

(3) 二次型的规范形为 z_1^2 .

(4) A 的特征值为 0, 0, 3, 则矩阵 $A - tI$ 的特征值为 $-t, -t, 3 - t$

当 $A - tI$ 为正定矩阵时, 其特征值均大于 0, 即 $\begin{cases} -t > 0 \\ 3 - t > 0 \end{cases}$

所以 $t < 0$.